SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1105

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 60%

QUESTIONS: 6 **Total Marks:60**

Annexure A

Application Based Questions

Code of Conduct:

I DARSHINIE KARANI.V.M (192520033) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Application Based Questions

1. Use Case Diagram Analysis

A **Smart Pharmacy Management System** allows customers to search medicines, upload prescriptions, place medicine orders, and make online payments. Pharmacists verify prescriptions and dispense medicines, while administrators manage inventory and user accounts. The current system design does not clearly define actor responsibilities and interactions.
Analyze the system requirements, identify all actors and missing use cases, and design an optimized **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to improve system functionality.
(10 Marks)

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2. Class Diagram Design

A **Smart Library Management System** consists of entities such as Book, Member, Librarian, Reservation, Fine, and BorrowRecord. The existing system does not clearly specify class attributes, operations, or relationships among these entities.
Analyze the system, identify relevant classes with appropriate attributes and methods, define relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a well-structured **Class Diagram**.
(10 Marks)

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3. Sequence Diagram Evaluation

An **Online Examination System** enables students to log in, select an examination, answer questions, submit responses, and receive results. The communication among the student, examination module, evaluation module, and database is incomplete.
Analyze the interaction process, identify all participating objects and missing message exchanges, and design a detailed **Sequence Diagram** representing the complete interaction flow with activation bars and lifelines.
(10 Marks)

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4. Activity Diagram Construction

A **Loan Approval System** allows customers to submit loan applications, upload supporting documents, complete eligibility verification, receive loan approval or rejection, and complete loan disbursement. The workflow lacks proper decision points and parallel activities.
Analyze the business process, identify all workflow activities, include decision nodes for approval and rejection, represent parallel processing where applicable, and construct an optimized **Activity Diagram**.
(10 Marks)

5. Component Diagram Design

A **Smart E-Commerce Platform** consists of modules such as User Management, Product Catalog, Shopping Cart, Order Management, Payment Gateway, Inventory Management, and Notification Service. The existing architecture does not clearly define interfaces and dependencies among modules.
Analyze the software architecture, identify major components and their provided/required interfaces, define dependencies among components, and design a comprehensive **Component Diagram** representing the overall system architecture.
(10 Marks)

6. Deployment Diagram Design

A **Smart Agriculture Monitoring System** uses IoT soil sensors, weather stations, irrigation controllers, cloud servers, mobile applications, and farmer dashboards to monitor crop health and automate irrigation. The deployment architecture lacks proper representation of hardware nodes and communication links.
Analyze the deployment environment, identify hardware nodes and deployed software artifacts, specify communication protocols between nodes, and construct a detailed **Deployment Diagram** representing the complete physical system architecture.
(10 Marks)

RUBRICS FOR CALCULATION

Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate	Applies relevant concepts	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
		concepts to solve complex problems	correctly with minor errors		
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Answers

1. **Actors**

1. Customer
2. Pharmacist
3. Administrator
4. Payment Gateway (External System)

Use Cases

- Register
- Login
- Search Medicine
- View Medicine Details
- Upload Prescription
- Verify Prescription
- Place Order
- Make Payment
- Generate Invoice
- Track Order
- Dispense Medicine
- Manage Inventory
- Manage Users
- Update Stock
- Logout

Relationships

<<include>>

- Place Order → Search Medicine
- Place Order → Upload Prescription
- Place Order → Make Payment
- Manage Inventory → Update Stock
- Make Payment → Generate Invoice

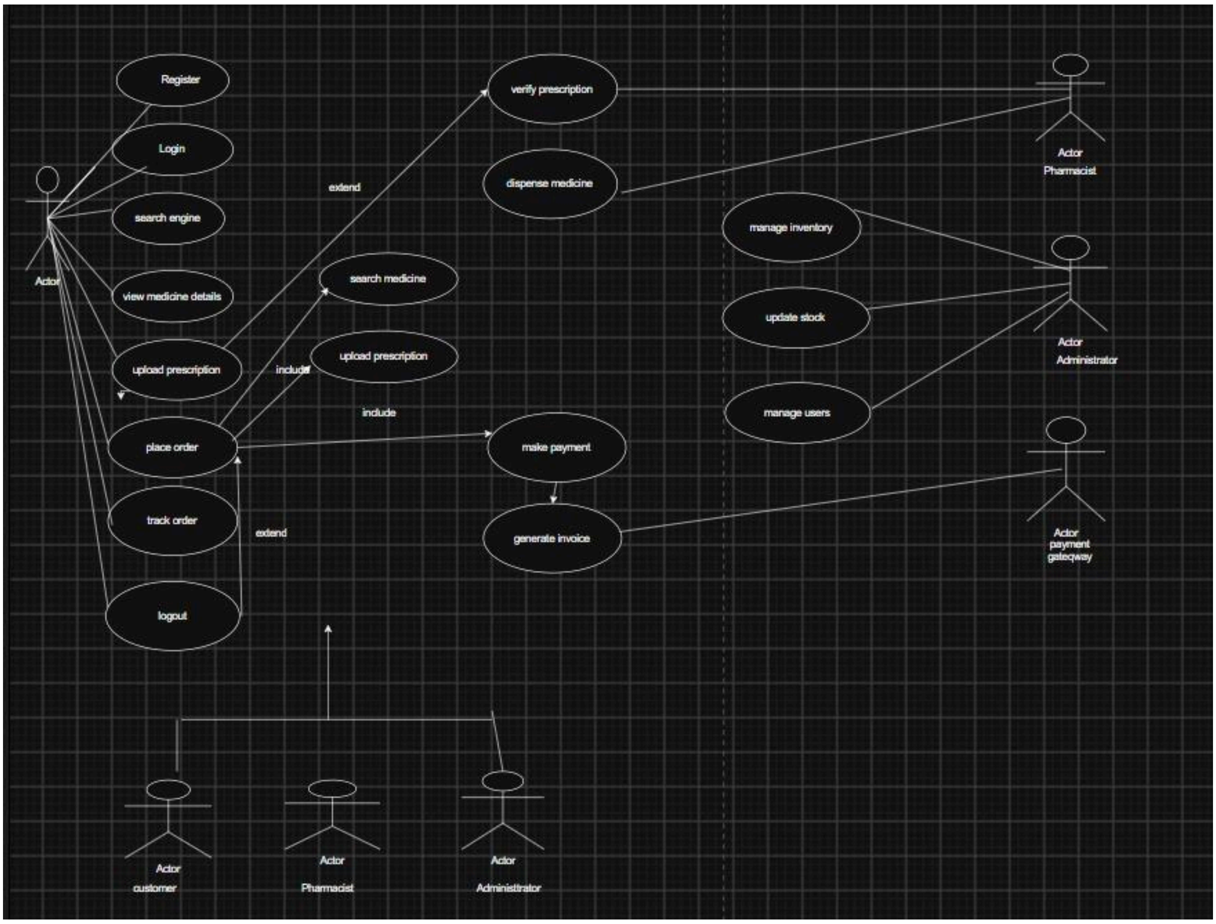
<<extend>>

- Verify Prescription → Place Order (only for prescription medicines)
- Track Order → Place Order

Generalization

- User
  - Customer
  - Pharmacist
  - Administrator

Diagram:



2.

The Smart Library Management System helps manage books, members, librarians, reservations, borrowing records, and fines. Each entity has specific responsibilities and interacts with other entities. A class diagram represents these classes, their attributes, methods, relationships, and multiplicities.

Classes with Attributes and Methods

1. User (Superclass)

Attributes:

- userID : int
- name : String
- email : String
- phoneNo : String

Methods:

- login()

- logout()

2. Member (Subclass of User)

Attributes:

- memberID : int
- membershipType : String

Methods:

- borrowBook()
- returnBook()
- reserveBook()

3. Librarian (Subclass of User)

Attributes:

- librarianID : int
- department : String

Methods:

- addBook()
- removeBook()
- issueBook()
- collectFine()

4. Book

Attributes:

- bookID : int
- title : String
- author : String
- category : String
- availability : Boolean

Methods:

- searchBook()
- updateStatus()

5. Reservation

Attributes:

- reservationID : int

- reservationDate : Date
- status : String

Methods:

- reserveBook()
- cancelReservation()

6. BorrowRecord

Attributes:

- borrowID : int
- issueDate : Date
- dueDate : Date
- returnDate : Date

Methods:

- calculateDueDate()
- updateReturnDate()

7. Fine

Attributes:

- fineID : int
- amount : double
- paymentStatus : String

Methods:

- calculateFine()
- payFine()

3. Relationships

Association:

- Member ↔ BorrowRecord
- BorrowRecord ↔ Book
- Member ↔ Reservation

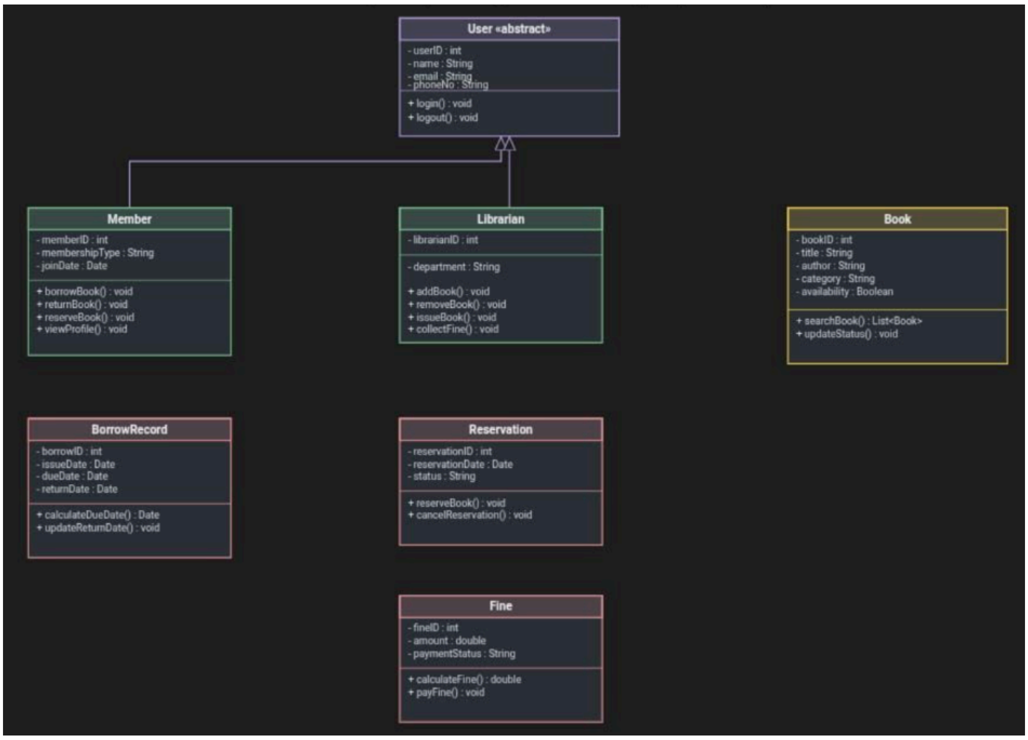
Aggregation (◇):

- Librarian ◇── Book
- *(A librarian manages books, but books can exist independently.)*

Composition (◆):

- BorrowRecord ◆── Fine
- *(A fine exists only if there is a borrow record.)*

<u>Diagram:</u>



Explanation

- User is the parent class.
- Member and Librarian inherit common properties from User.
- A Member can borrow many books and make multiple reservations.
- Each borrowing transaction is stored in a BorrowRecord.
- A Fine is associated with a borrow record and exists only when required.
- A Librarian manages books in the library but does not own them, so aggregation is used.

3.

The Online Examination System allows students to take examinations online. The system verifies the student's login credentials, loads the selected examination, stores answers, evaluates them, saves the result, and finally displays the score.

<u>Participating Objects (Lifelines)</u>

1. Student
2. Login Page
3. Examination Module

4. Evaluation Module
5. Database

<u>Message Exchanges</u>

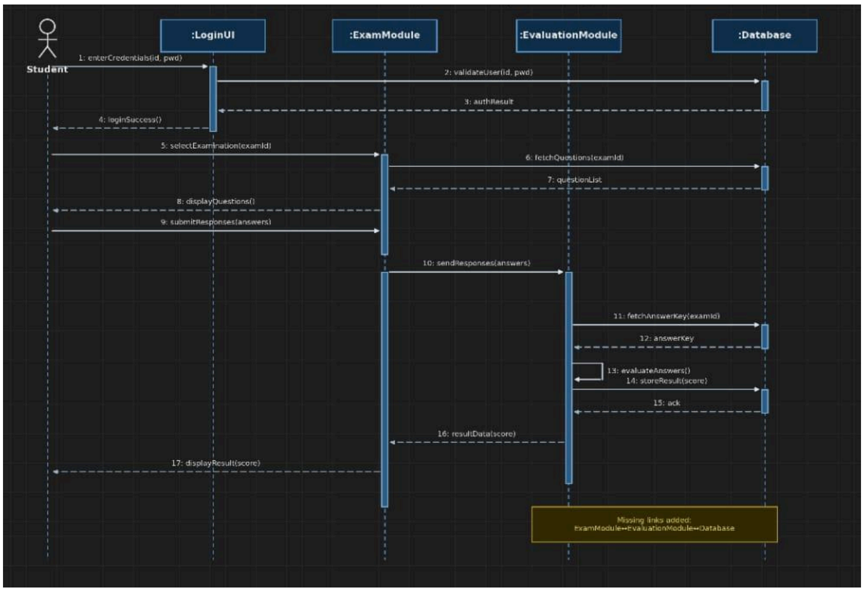
1. Student enters username and password.
2. Login Page sends credentials to Database.
3. Database validates credentials.
4. Login successful message is returned.
5. Student selects an examination.
6. Examination Module requests questions from Database.
7. Database returns question paper.
8. Student answers questions.
9. Student submits answers.
10. Examination Module stores responses in Database.
11. Examination Module sends answers to Evaluation Module.
12. Evaluation Module evaluates the answers.
13. Evaluation Module stores marks in Database.
14. Database confirms storage.
15. Evaluation Module sends result to Student.

<u>Activation Bars:</u>

Activation bars should be drawn on the following objects while they are executing operations:

- Login Page – during login verification.
- Database – while validating login, retrieving questions, storing answers, and saving results.
- Examination Module – while loading questions and submitting answers.
- Evaluation Module – while evaluating answers and generating results.

<u>Diagram:</u>



Explanation

- The Student initiates the examination process by logging in.
- The Login Page validates the credentials using the Database.
- After successful authentication, the Examination Module retrieves the selected examination questions.
- The student answers the questions and submits them.
- The Evaluation Module evaluates the submitted answers and stores the marks in the Database.
- Finally, the system displays the examination result to the student.

4.

The Loan Approval System is designed to process loan applications submitted by customers. The system verifies customer details and uploaded documents, performs parallel verification processes, decides whether the loan is approved or rejected, and if approved, disburses the loan amount.

<u>Workflow Activities</u>

1. Start
2. Customer Login
3. Submit Loan Application
4. Upload Supporting Documents
5. Verify Application Details
6. Fork (Parallel Activities)
    - Income Verification
    - Document Verification
7. Join
8. Eligibility Verification
9. Decision Node
    - Eligible?
10. Loan Approved / Loan Rejected
11. Loan Disbursement (if approved)
12. Notify Customer
13. End

<u>Decision Nodes</u>

Decision 1: Eligibility Verification

Yes:-

- Approve Loan
- Disburse Loan
- Notify Customer

No:-

- Reject Loan

- Notify Customer

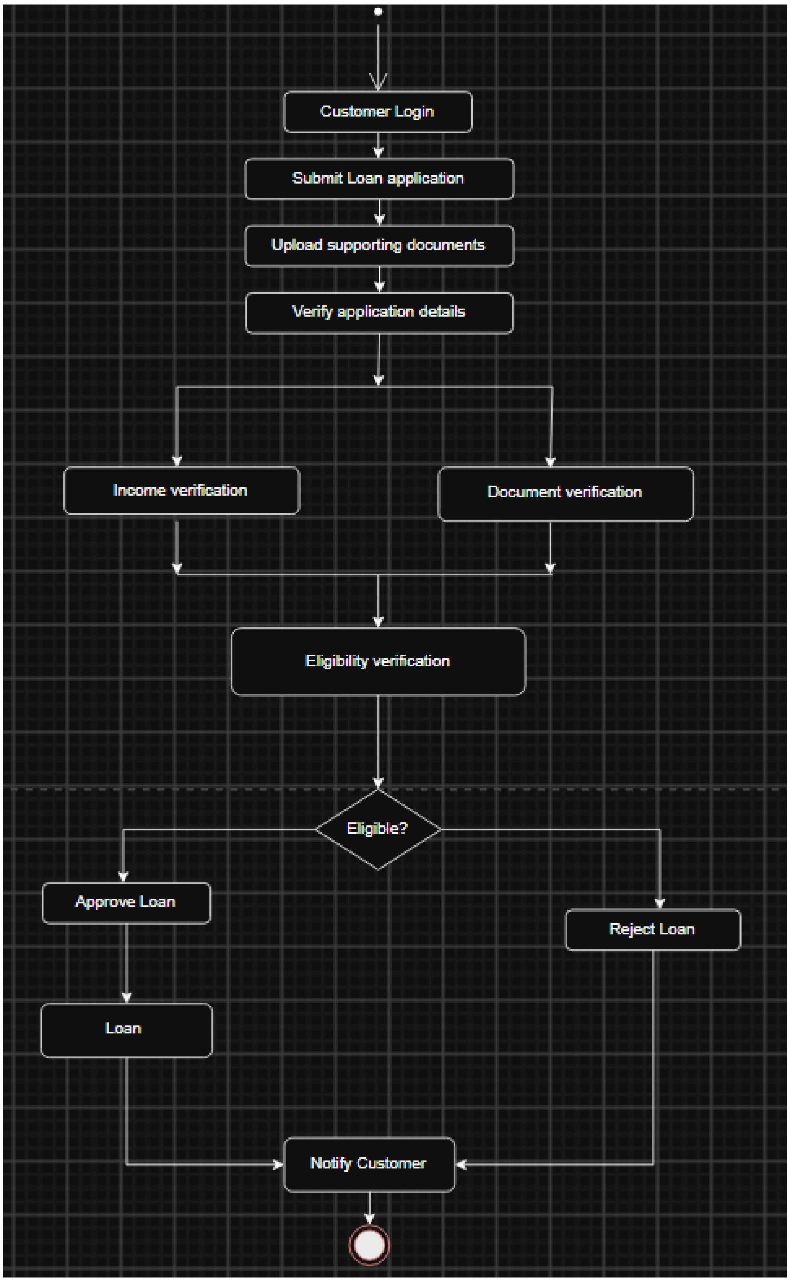
Parallel Activities

After verifying the application, the following activities occur simultaneously:

- Income Verification
- Document Verification

These two activities are represented using a Fork Node and are synchronized using a Join Node before checking eligibility.

Diagram:



Explanation:

- The customer logs in and submits the loan application along with the required documents.
- The system verifies the application details.
- Income Verification and Document Verification are performed in parallel using a Fork Node.
- After both processes are completed, they merge at the Join Node.
- The system checks the customer's eligibility using a Decision Node.
- If the customer is eligible, the loan is approved, the amount is disbursed, and the customer is notified.
- If the customer is not eligible, the loan is rejected, and the customer receives a rejection notification.
- The process then ends.

5.

The Smart E-Commerce Platform allows users to browse products, add items to the shopping cart, place orders, make online payments, manage inventory, and receive notifications. Each module performs a specific function and communicates with other modules through interfaces.

Major Components

1. User Management
2. Product Catalog
3. Shopping Cart
4. Order Management
5. Payment Gateway
6. Inventory Management
7. Notification Service
8. Database

Provided and Required Interfaces :

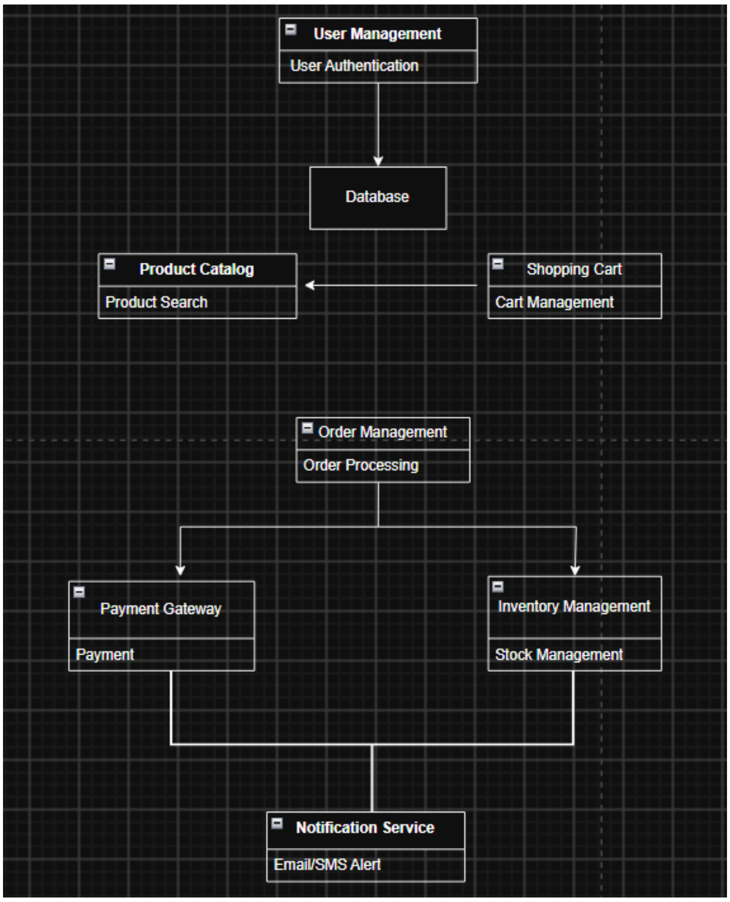
Component	Provided Interface	Required Interface
User Management	User Authentication	Database
Product Catalog	Product Search	Database
Shopping Cart	Cart Management	Product Catalog

Order Management	Order Processing	Shopping Cart, Payment Gateway, Inventory Management
Payment Gateway	Payment Service	Bank/Payment System
Inventory Management	Stock Management	Database
Notification Service	Email/SMS Notifications	Order Management
Database	Data Storage	—

Dependencies Among Components

- Shopping Cart depends on Product Catalog to display product details.
- Order Management depends on:
  - Shopping Cart
  - Payment Gateway
  - Inventory Management
- Inventory Management updates product stock in the Database.
- Notification Service depends on Order Management to send order confirmation and delivery updates.
- User Management authenticates users using the Database.

<u>Diagram:</u>



Explanation:

- User Management handles user registration and login by interacting with the Database.
- Product Catalog provides product details and search functionality.
- Shopping Cart stores the products selected by the user.
- Order Management processes customer orders by communicating with the Shopping Cart, Payment Gateway, and Inventory Management.
- Payment Gateway securely processes online payments.
- Inventory Management updates stock availability after successful orders.
- Notification Service sends order confirmation and delivery notifications to customers.

6.

The Smart Agriculture Monitoring System uses IoT devices to collect real-time data such as soil moisture and weather conditions. The collected data is sent to a cloud server, where it is processed and stored. Farmers can monitor crop health through a mobile application or web dashboard, and the system automatically controls irrigation based on sensor data.

Hardware Nodes:

1. IoT Soil Sensor
2. Weather Station
3. Irrigation Controller

4. Cloud Server
5. Database Server
6. Farmer Mobile Device
7. Farmer Dashboard (PC/Laptop)

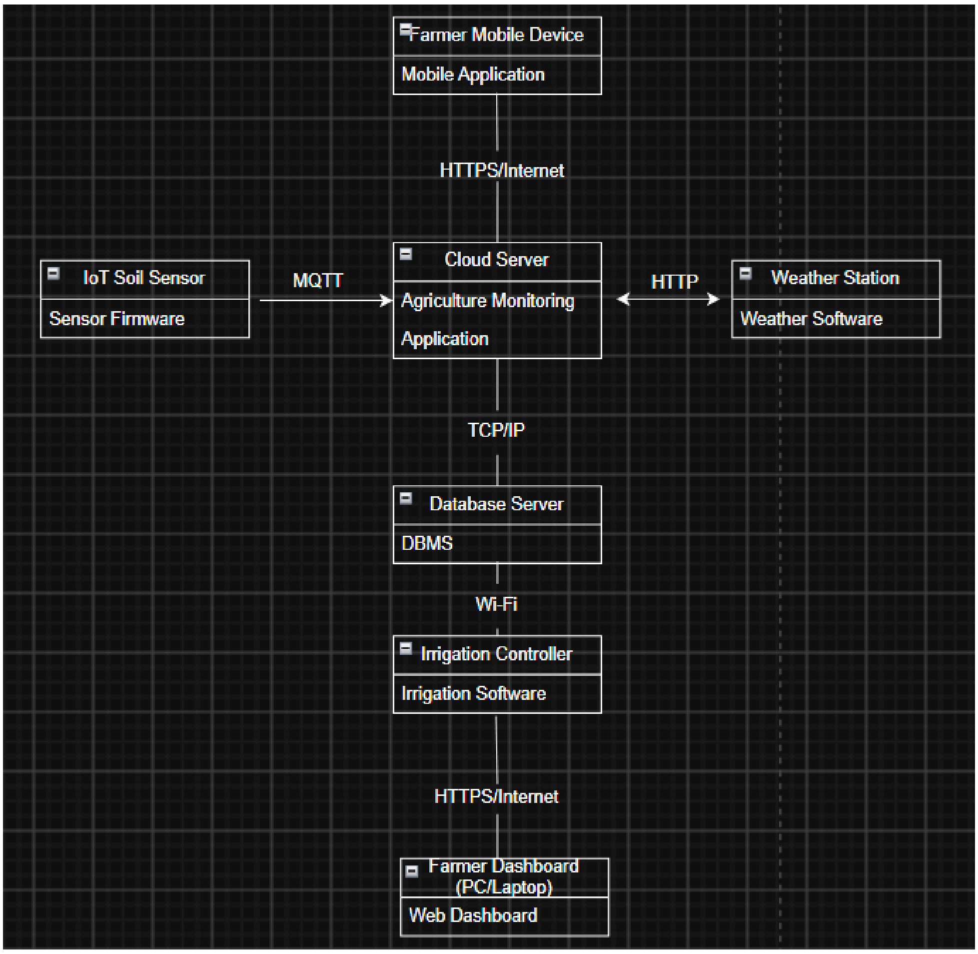
Software Artifacts:

Hardware Node	Deployed Software Artifact
IoT Soil Sensor	Sensor Firmware
Weather Station	Weather Monitoring Software
Irrigation Controller	Irrigation Control Software
Cloud Server	Agriculture Monitoring Application
Database Server	Database Management System (DBMS)
Farmer Mobile Device	Mobile Application
Farmer Dashboard	Web Dashboard

Communication Protocols:

- IoT Soil Sensor → Cloud Server : Wi-Fi / MQTT
- Weather Station → Cloud Server : Internet / HTTP
- Cloud Server → Database Server : TCP/IP
- Cloud Server → Irrigation Controller : Wi-Fi
- Cloud Server → Mobile Application : Internet / HTTPS
- Cloud Server → Farmer Dashboard : Internet / HTTPS

Diagram:



<u>Explanation:</u>

- IoT Soil Sensors collect soil moisture and transmit the data to the Cloud Server using MQTT over Wi-Fi.
- The Weather Station sends weather information to the Cloud Server through the Internet.
- The Cloud Server processes and stores all agricultural data in the Database Server.
- Based on the processed data, the Cloud Server sends commands to the Irrigation Controller to automate watering.
- Farmers can monitor crop health and irrigation status using the Mobile Application or Web Dashboard.